

NAME:KARTHIK
ROLL NO:241001107
WEEK-1

Overview of C, Constants, Variables and Data Types

Ex.No.: 1

Date: 4.10.24

Say "Hello, World!" With C

Problem Statement:

This is a simple challenge to help you practice printing to stdout.

We're starting out by printing the most famous computing phrase of all time! In the editor below, use either `printf` or `cout` to print the string `Hello, World!` to stdout.

Input Format

You do not need to read any input in this challenge.

Output Format

Print `Hello, World!` to stdout.

Sample Output 1

Hello, World!

Program:

```
1 #include<stdio.h>
2 int main()
3 {
4     printf("Hello, World!");
5     return 0;
6 }
```

	Expected	Got	
✓	Hello, World!	Hello, World!	✓

Passed all tests! ✓

Ex.No.: 2

Date: 4.10.24

Playing with Characters

Problem Statement:

This challenge will help you to learn how to take a character, a string and a sentence as input in C. To take a single character `ch` as input, you can use `scanf("%c", &ch);` and `printf("%c", ch)` writes a character specified by the argument `char` to `stdout`:

```
char ch;
scanf("%c", &ch);
printf("%c", ch);
```

This piece of code prints the character `ch`. You can take a string as input in C using `scanf("%s", s)`. But it accepts string only until it finds the first space. In order to take a line as input, you can use `scanf("%[^\n] %*c", s);` where `s` is defined as `chars [MAX_LEN]` where `MAX_LEN` is the maximum size of `s`. Here, `[]` is the scanset character. `^\n` stands for taking input until a newline isn't encountered. Then, with this `%*c`, it reads the newline character and here, the used `*` indicates that this newline character is discarded.

Note: After inputting the character and the string, inputting the sentence by the above mentioned statement won't work. This is because, at the end of each line, a new line character (`\n`) is present. So, the statement: `scanf("%[^\n] %*c", s);` will not work because the last statement will read a newline character from the previous line. This can be handled in a variety of ways and one of them being: `scanf("\n");` before the last statement. Task: You have to print the character, `ch`, in the first line. Then print `s` in next line. In the last line print the sentence, `sen`.
Input Format

First, take a character, `ch` as input. Then take the string, `s` as input. Lastly, take the sentence `sen` as input

Output Format

Print three lines of output. The first line prints the character, `ch`. The second line prints the string, `s`. The third line prints the sentence, `sen`.

Sample Input 1

```
C
program
Programming using C
```

Sample Output 1

```
C
program
Programming using C
```

Program:

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     char ch;
5     scanf("%c",&ch);
6     printf("%c",ch);
7     return 0;
8 }
```

	Input	Expected	Got	
✓	C	C	C	✓

Passed all tests! ✓

Sum and Difference of Two Numbers

Problem Statement:

The fundamental data types in C are int, float and char. Today, we're discussing int and float data types.

The printf() function prints the given statement to the console. The syntax is printf("format string", argument_list);. In the function, if we are using an integer, character, string or float as argument, then in the format string we have to write %d (integer), %c (character), %s (string), %f (float) respectively.

The scanf() function reads the input data from the console. The syntax is scanf("format string", argument_list);. For ex: The scanf("%d",&number) statement reads integer number from the console and stores the given value in variable number. To input two integers separated by a space on a single line, the command is scanf("%d %d", &n, &m), where n and m are the two integers.

Task

Your task is to take two numbers of int data type, two numbers of float data type as input and output their sum:

1. Declare 4 variables: two of type int and two of type float.
2. Read 2 lines of input from stdin (according to the sequence given in the 'Input Format'

section below) and initialize your 4 variables.

3. Use the + and - operator to perform the following operations:
Print the sum and difference of two int variable on a new line.
Print the sum and difference of two float variable rounded to one decimal place on a new line.

Input Format

The first line contains two integers. The second line contains two floating point numbers.

Constraints: $1 \leq \text{integer variables} \leq 104$, $1 \leq \text{float variables} \leq 104$

Output Format

Print the sum and difference of both integers separated by a space on the first line, and the sum and difference of both float (scaled to 1 decimal place) separated by a space on the second line.

Sample Input

10 4

4.0 2.0

Sample Output

14 6

6.0 2.0

Program:

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main()
3  {
4      int a,b;
5      float c,d;
6      scanf("%d%d",&a,&b);
7      scanf("%f%f",&c,&d);
8      printf("%d %d\n",a+b,a-b);
9      printf("%.1f %.1f",c+d,c-d);
10     return 0;
11 }
12
13
14
15

```

	Input	Expected	Got	
✓	10 4 4.0 2.0	14 6 6.0 2.0	14 6 6.0 2.0	✓
✓	20 8 8.0 4.0	28 12 12.0 4.0	28 12 12.0 4.0	✓

Passed all tests! ✓

Ex.No.: 4

Date: 4.10.24

Average Marks

Problem Statement

Write a program to input a name (as a single character) and marks of three tests as m1, m2, and m3 of a student considering all the three marks have been given in integer format.

Now, you need to calculate the average of the given marks and print it along with the name as mentioned in the output format section.

All the test marks are in integers and hence calculate the average in integer as well. That is, you need to print the integer part of the average only and neglect the decimal part.

Input Format :

Line 1 : Name(Single character)

Line 2 : Marks scored in the 3 tests separated by single space.

Output Format:

First line of output prints the name of the student. Second line of the output prints the average mark.

Constraints

Marks for each student lie in the range 0 to 100 (both inclusive)

Sample Input 1 :

A

3 4 6

Sample Output 1 :

A
4

Program:

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main()
3  {
4      char a;
5      int b,c,d;
6      scanf("%c",&a);
7      scanf("%d %d %d",&b,&c,&d);
8      printf("%c\n",a);
9      printf("%d ",(b+c+d)/3);
10     return 0;
11 }
12
13

```

	Input	Expected	Got	
✓	A 3 4 6	A 4	A 4	✓
✓	T 7 3 8	T 6	T 6	✓
✓	R 0 100 99	R 66	R 66	✓

Passed all tests! ✓

Ex.No.: 5

Date: 4.10.24

Basic Data Types

Problem Statement:

Some C data types, their format specifiers, and their most common bit widths are as follows:

Int ("%d"): 32 Bit integer
 Long ("%ld"): 64 bit integer
 Char ("%c"): Character type
 Float ("%f"): 32 bit real value
 Double ("%lf"): 64 bit real value

Reading

To read a data type, use the following syntax: `scanf("`format_specifier`", &val)`

For example, to read a character followed by a double: `char ch;`

`double d;`

`scanf("%c %lf", &ch, &d);`

For the moment, we can ignore the spacing between format specifiers.

Printing

To print a data type, use the following syntax: `printf("`format_specifier`", val)`

For example, to print a character followed by a double: `char ch = 'd';`

`double d = 234.432;`

`printf("%c %lf", ch, d);`

Note: You can also use `cin` and `cout` instead of `scanf` and `printf`; however, if you are taking a million numbers as input and printing a million lines, it is faster to use `scanf` and `printf`.

Input Format

Input consists of the following space-separated values: `int`, `long`, `char`, `float`, and `double`, respectively.

Output Format

Print each element on a new line in the same order it was received as input. Note that the floating-point value should be correct up to 3 decimal places and the double to 9 decimal places.

Sample Input

```
3
12345678912345
a
334.23
14049.30493
```

Sample Output

```
3
12345678912345
a
334.230
14049.304930000
```

Program:

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main()
3  {
4      int a;
5      long b;
6      char c;
7      float d;
8      double e;
9      scanf("%d %ld %c %f %lf",&a,&b,&c,&d,&e);
10     printf("%d\n%ld\n%c\n%.3f\n%.9lf",a,b,c,d,e);
11     return 0;
12 }
```

	Input	Expected	Got	
✓	3 12345678912345 a 334.23 14049.30493	3 12345678912345 a 334.230 14049.304930000	3 12345678912345 a 334.230 14049.304930000	✓

Passed all tests! ✓

Ex.No.: 6

Date: 4.10.24

ASCII Value and Adjacent Characters

Problem Statement:

Write a program to print the ASCII value and the two adjacent characters of the given character.

Input Format: Reads the character

Output Format: First line prints the ascii value, second line prints the previous character and next character of the input character

Sample Input 1:

E

Sample Output 1:

69

D F

Program:

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main()
3  {
4      char a;
5      scanf("%c",&a);
6      printf("%d\n",a);
7      printf("%c %c",a-1,a+1);
8      return 0;
9
10 }
```

	Input	Expected	Got	
✓	E	69 D F	69 D F	✓

Passed all tests! ✓